Multi-access Edge Computing (MEC): Framework and Reference Architecture

Based on: ETSI GS MEC 003 V3.2.1 (2024-04). Multi-access Edge Computing (MEC); Framework and Reference Architecture. ETSI Industry Specification Group (ISG) MEC.

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Introduction

The rapid growth of 5G applications such as autonomous vehicles, augmented reality (AR), industrial automation, smart cities, and real-time video analytics has created a need for extremely low communication and processing delays. Conventional cloud computing sends user data to centralized data centres, which can introduce significant network latency and consume additional bandwidth. Multi-access Edge Computing (MEC) addresses this limitation by moving computing, storage, and application services closer to the users at the edge of the network. According to the ETSI MEC Framework and Reference Architecture (GS MEC 003), MEC provides cloud-computing capabilities at the network edge with ultra-low latency, high bandwidth, and access to real-time network information. This makes MEC an important enabling technology for demanding 5G applications.

MEC Architecture

A typical MEC architecture consists of the 5G User Equipment (UE), Radio Access Network (RAN), MEC host, MEC platform, MEC applications, and MEC orchestrator. The UE generates application data that is transmitted through the 5G RAN. Instead of forwarding all data to a distant centralized cloud, selected processing tasks can be executed on a nearby MEC host. The MEC host contains the virtualization infrastructure, MEC platform, and MEC applications. The MEC platform provides common services required by applications and enables them to access relevant network information. The MEC orchestrator performs higher-level management, including coordinating MEC resources and supporting application deployment and lifecycle management.

Role in Low-Latency 5G Applications

The main advantage of MEC is the reduction of end-to-end latency. For example, in an autonomous vehicle system, sensor information can be processed by a nearby edge server instead of being transmitted to a remote cloud. This enables faster decisions for applications such as collision avoidance and traffic coordination. Similarly, industrial robots can exchange control information with nearby edge computing nodes, reducing delays that could affect real-time operation. ETSI also identifies use cases such as V2X, industrial IoT, and other latency-sensitive applications where MEC can provide significant benefits. MEC can additionally reduce backhaul traffic because data can be processed locally instead of continuously being sent to centralized cloud infrastructure.

Technologies Used

Although MEC provides significant performance improvements, deploying distributed edge infrastructure introduces challenges. Mobility is another challenge because a user may move between different MEC coverage areas, requiring applications or processing tasks to maintain service continuity. Security, privacy, interoperability, and management of numerous distributed edge nodes are also important considerations. ETSI's work on MEC and 5G integration addresses interactions between MEC applications, the 5G Core Network, and the RAN, demonstrating that MEC is not simply an independent computing platform but must be closely integrated with the 5G network architecture.

Conclusion

MEC is a key architecture for realizing the low-latency and high-bandwidth objectives of 5G. By moving computation closer to users, it reduces communication delays, lowers backhaul traffic, and enables real-time applications that are difficult to support using centralized cloud computing alone. The ETSI MEC reference architecture provides a standardized framework consisting of MEC hosts, platforms, applications, and orchestration functions. Therefore, MEC combined with 5G provides a strong foundation for future applications such as autonomous transportation, smart manufacturing, immersive services, and intelligent IoT systems.

ETSI GS MEC 003 V3.2.1 (2024-04)

Multi-access Edge Computing (MEC); Framework and Reference Architecture

Disclaimer

Reference

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Keywords

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Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Multi-access Edge Computing (MEC).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document provides a framework and reference architecture for Multi-access Edge Computing that describes a MEC system that enables MEC applications to run efficiently and seamlessly in a multi-access network. The present document also describes the functional elements and the reference points between them, and a number of MEC services that comprise the solution. It finally presents a number of key concepts related to the multi-access edge architecture.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference/.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI GS MEC 002: "Multi-access Edge Computing (MEC); Use Cases and Requirements".

[2] ETSI GS NFV 002: "Network Functions Virtualisation (NFV); Architectural Framework".

[3] ETSI GS NFV-IFA 013: "Network Functions Virtualisation (NFV); Management and Orchestration; Os-Ma-nfvo reference point - Interface and Information Model Specification".

[4] ETSI GS NFV-IFA 008: "Network Functions Virtualisation (NFV); Management and Orchestration; Ve-Vnfm reference point - Interface and Information Model Specification".

[5] ETSI GS MEC 009: "Multi-access Edge Computing (MEC); General principles, patterns and common aspects of MEC Service APIs".

[6] ETSI GS MEC 040: "Multi-access Edge Computing (MEC); Federation enablement APIs".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ETSI GR MEC 001: "Multi-access Edge Computing (MEC); Terminology".

[i.2] Void.

[i.3] OpenStack®: "OpenStack++ for Cloudlet Deployment".

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[i.4] Void.

[i.5] ETSI GR MEC 017: "Mobile Edge Computing (MEC); Deployment of Mobile Edge Computing in an NFV environment".

[i.6] ETSI TS 123 501: "5G; System architecture for the 5G System (5GS) (3GPP TS 23.501 Release 17)".

[i.7] ETSI GR MEC 035: "Multi-access Edge Computing (MEC); Study on Inter-MEC systems and MEC-Cloud systems coordination".

[i.8] GSMA Official Document OPG.02: "Operator Platform: Requirements and Architecture", v6.0, February 2024.

[i.9] ETSI TS 123 558: "5G; Architecture for enabling Edge Applications (3GPP TS 23.558 Release 17)".

[i.10] ETSI TS 133 310: "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Network Domain Security (NDS); Authentication Framework (AF) (3GPP TS 33.310 Release 17)".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI GR MEC 001 [i.1] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI GR MEC 001 [i.1] and the following apply:

CFS Customer Facing Service

4 Overview

The present document presents a framework and a reference architecture to support the requirements defined for Multi-access Edge Computing in ETSI GS MEC 002 [1].

The framework described in clause 5 shows the structure of the Multi-access Edge Computing environment.

The reference architecture described in clause 6 shows the functional elements that compose the multi-access edge system, including the MEC platform and the MEC management, as well as the reference points between them.

The functional elements and reference points listed in clause 7 describe the high-level functionality of the different functional elements and reference points.

Clause 8 describes the high-level functionality of a number of MEC services, comprising the solution for Multi-access Edge Computing.

Annex A describes at a high-level a number of key concepts that underlie the principles used to develop the framework and reference architecture described in the present document.

5 Multi-access Edge Computing framework

Multi-access Edge Computing enables the implementation of MEC applications as software-only entities that run on top of a Virtualisation infrastructure, which is located in or close to the network edge. The Multi-access Edge Computing framework shows the general entities involved. These can be grouped into system level, host level and network level entities.

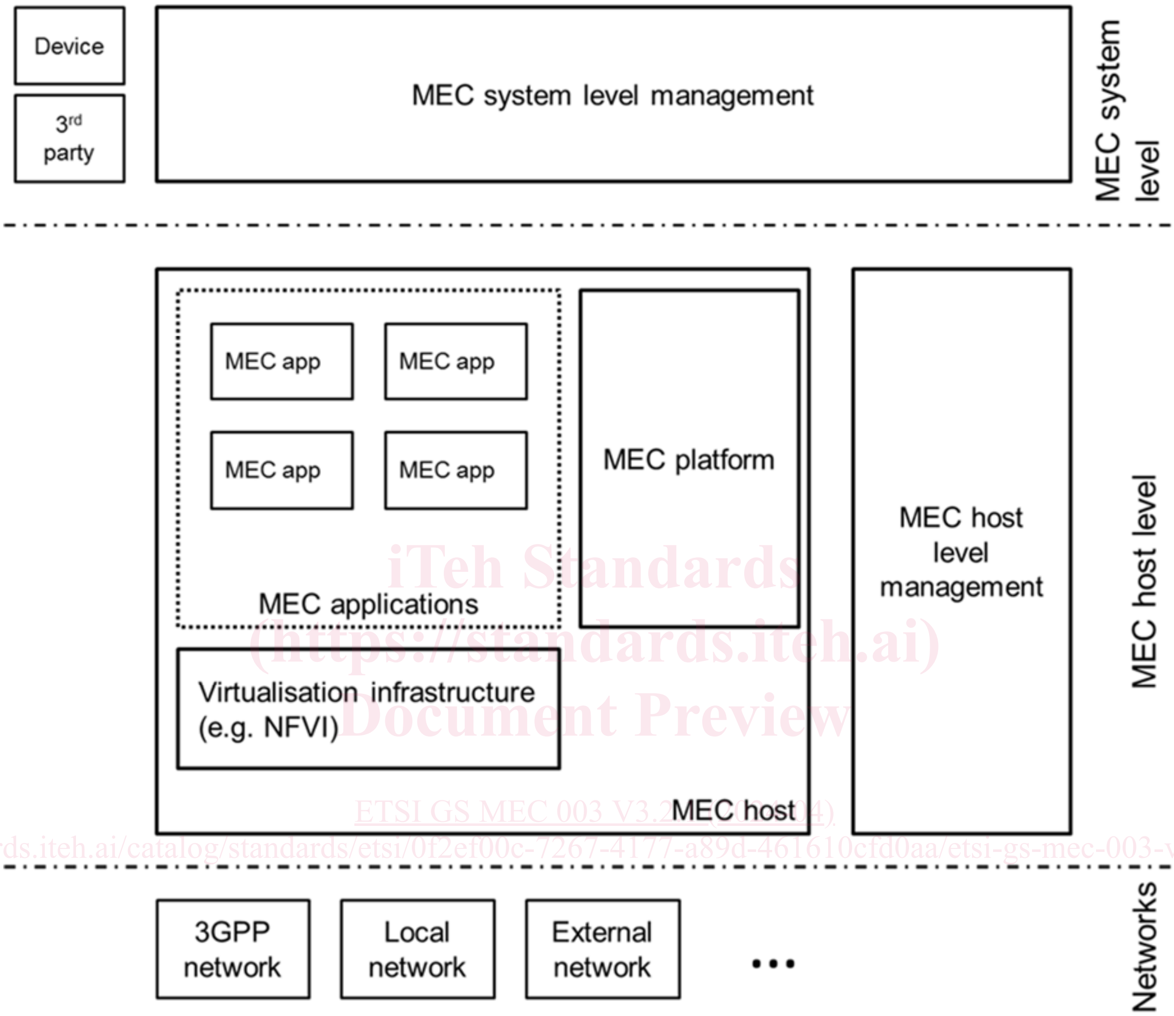

Figure 5-1: Multi-access Edge Computing framework

Figure 5-1 illustrates the framework for Multi-access Edge Computing consisting of the following entities:

- MEC host, including the following:
  - MEC platform;
  - MEC applications;
  - virtualisation infrastructure;
- MEC system level management;
- MEC host level management;
- external related entities, i.e. network level entities.

6 Reference architecture

6.1 Generic reference architecture

The reference architecture shows the functional elements that comprise the multi-access edge system and the reference points between them.

Figure 6-1 depicts the generic multi-access edge system reference architecture. There are three groups of reference points defined between the system entities:

- reference points regarding the MEC platform functionality (Mp);
- management reference points (Mm); and
- reference points connecting to external entities (Mx).

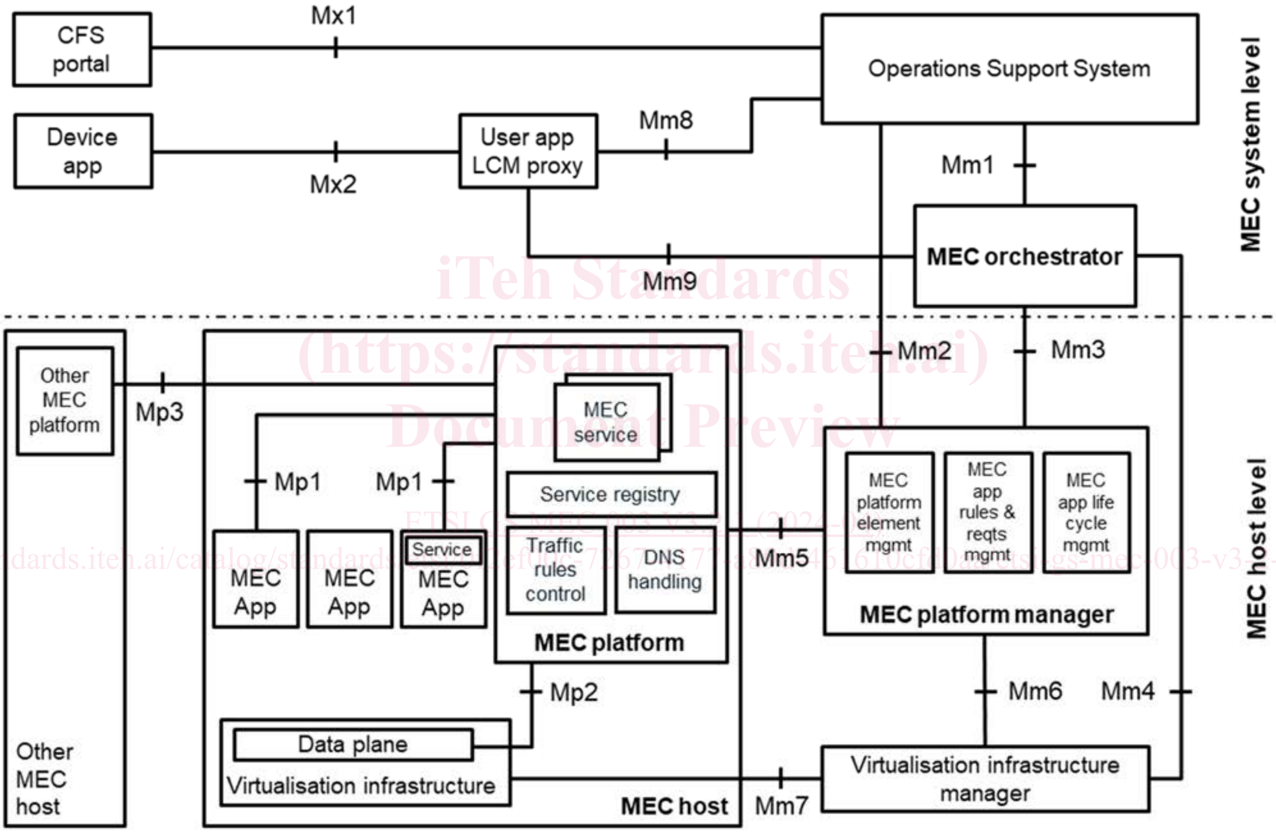

Figure 6-1: Multi-access edge system reference architecture

The multi-access edge system consists of the MEC hosts and the MEC management necessary to run MEC applications within an operator network or a subset of an operator network.

The **MEC host** is an entity that contains a MEC platform and a Virtualisation infrastructure which provides compute, storage, and network resources, for the purpose of running MEC applications. The MEC host is further described in clause 7.1.1.

The **MEC platform** is the collection of essential functionality required to run MEC applications on a particular Virtualisation infrastructure and enable them to provide and consume MEC services. The MEC platform can also provide services. The MEC platform is further described in clause 7.1.2.

MEC applications are instantiated and run on the Virtualisation infrastructure based on configuration or based on requests validated by the MEC management. MEC applications are further described in clause 7.1.3.

The MEC management comprises the MEC system level management and the MEC host level management.

The MEC system level management includes the **MEC orchestrator** as its core component, which has an overview of the complete MEC system. The MEC system level management is further described in clause 7.1.4.

The MEC host level management comprises the **MEC platform manager** and the **Virtualisation infrastructure manager**, and handles the management of the MEC specific functionality of a particular MEC host and the applications running on it. The MEC host level management is further described in clause 7.1.5.

6.2 Reference architecture variant for MEC in NFV

6.2.1 Description

MEC and Network Functions Virtualisation (NFV) are complementary concepts. The MEC architecture has been designed in such a way that a number of different deployment options of MEC systems are possible. A dedicated Group Report, ETSI GR MEC 017 [i.5], provides an analysis of solution details of the deployment of MEC in an NFV environment.

In clauses 6.2.2, 7.1.8 and 7.2.4 of the present document, a MEC architecture variant is specified that allows to instantiate MEC applications and NFV virtualised network functions on the same Virtualisation infrastructure, and to re-use ETSI NFV MANO components to fulfil a part of the MEC management and orchestration tasks.

6.2.2 Architecture diagram

Figure 6-2 depicts a variant of the multi-access edge system reference architecture for the deployment in an NFV environment [2].

In addition to the definitions for the generic reference architecture in clause 6.1, the following new architectural assumptions apply:

- The MEC platform is deployed as a VNF.
- The MEC applications appear as VNFs towards the ETSI NFV MANO components.
- The Virtualisation infrastructure is deployed as an NFVI and is managed by a VIM as defined by ETSI GS NFV 002 [2].
- The MEC Platform Manager (MEPM) is replaced by a MEC platform manager - NFV (MEPM-V) that delegates the VNF lifecycle management to one or more VNF Managers (VNFMs).
- The MEC Orchestrator (MEO) is replaced by a MEC Application Orchestrator (MEAO) that relies on the NFV Orchestrator (NFVO) for resource orchestration and for orchestration of the set of MEC application VNFs as one or more NFV Network Services (NSs).

The new reference points shown in figure 6-2 are further described in clause 7.2.4.